

Assignment 1: Java Tools and IDE, Simple Java Programs

Set A

- a) Using javap, view the methods of the following classes from the lang package: java.lang.Object, java.lang.String and java.util.Scanner. and also Compile sample program 8. Type the following command and view the bytecodes. `javap -c MyClass`.
- b) Write a program to calculate perimeter and area of rectangle. (hint : area = length * breadth, perimeter=2*(length+breadth))
- c) Write a menu driven program to perform the following operations
 - i. Calculate the volume of cylinder. (hint : Volume: $\pi \times r^2 \times h$)
 - ii. Find the factorial of given number.
 - iii. Check the number is Armstrong or not.
 - iv. Exit
- d) Write a program to accept the array element and display in reverse order.

Set B)

- a) Write a java program to display the system date and time in various formats shown below
Current date is : 31/08/2021
Current date is : 08-31-2021
Current date is : Tuesday August 31 2021
Current date and time is : Fri August 31 15:25:59 IST 2021
Current date and time is : 31/08/21 15:25:59 PM +0530 Current time is : 15:25:59 Current week of year is : 35 Current week of month : 5 Current day of the year is : 243 Note: Use java.util.Date and java.text.SimpleDateFormat class
- b) Define a class MyNumber having one private int data member. Write a default constructor to initialize it to 0 and another constructor to initialize it to a value (Use this). Write methods isNegative, isPositive, isZero, isOdd, isEven. Create an object in main. Use command line arguments to pass a value to the object (Hint : convert string argument to integer) and perform the above tests. Provide javadoc comments for all constructors and methods and generate the html help file.
- c) Write a menu driven program to perform the following operations on multidimensional array ie matrix : i. Addition ii. iii. iv. Multiplication Transpose of any matrix. Exit

Set C)

- a) Write a program to accept n names of country and display them in descending order.
- b) Write a menu driven program to perform the following operations on 2D array: i. Sum of diagonal elements ii. iii. iv. Sum of upper diagonal elements Sum of lower diagonal elements Exit
- c) Write a program to display the 1 to 15 tables. ($1 * 1 = 1$ $2 * 1 = 2$ $1 * 2 = 2$ $1 * 3 = 3$... $1 * 10 = 10$ Assignment Evaluation 0: Not Done $2 * 2 = 4$ $2 * 3 = 6$... $2 * 10 = 20$ $15 * 1 = 15$ $15 * 2 = 30$ $15 * 3 = 45$ $15 * 10 = 150$)

Assignment 2: Array of Objects and Packages

Set A)

a) Create an employee class(id,name,deptname,salary). Define a default and parameterized constructor. Use 'this' keyword to initialize instance variables. Keep a count of objects created. Create objects using parameterized constructor and display the object count after each object is created.(Use static member and method). Also display the contents of each object.

b) Define Student class(roll_no, name, percentage) to create n objects of the Student class. Accept details from the user for each object. Define a static method "sortStudent" which sorts the array on the basis of percentage.

c) Write a java program to accept 5 numbers using command line arguments sort and display them.

d) Write a java program that take input as a person name in the format of first, middle and last name and then print it in the form last, first and middle name, where in the middle name first character is capital letter.

Set B

a) Write a Java program to create a Package "SY" which has a class SYMarks (members – ComputerTotal, MathsTotal, and ElectronicsTotal). Create another package TY which has a class TYMarks (members – Theory, Practicals). Create n objects of Student class (having rollNumber, name, SYMarks and TYMarks). Add the marks of SY and TY computer subjects and calculate the Grade ('A' for ≥ 70 , 'B' for ≥ 60 'C' for ≥ 50 , Pass Class for ≥ 40 else 'FAIL') and display the result of the student in proper format.

b) Define a class CricketPlayer (name,no_of_innings,no_of_times_notout, totatruns, bat_avg). Create an array of n player objects .Calculate the batting average for each player using static method avg(). Define a static sort method

which sorts the array on the basis of average. Display the player details in sorted order.

Set C

a) Write a package for String operation which has two classes Con and Comp. Con class has to concatenate two strings and comp class compares two strings. Also display proper message on execution.

b) Create four member variables for Customer class. Assign public, private, protected and default access modifiers respectively to these variables. Try to access these variables from other classes (Same package and Different package)

Assignment 3: Inheritance and Interfaces

Set A)

- a) Write a program for multilevel inheritance such that country is inherited from continent. State is inherited from country. Display the place, state, country and continent.
- b) Define an abstract class Staff with protected members id and name. Define a parameterized constructor. Define one subclass OfficeStaff with member department. Create n objects of OfficeStaff and display all details.
- c) c) Define an interface "Operation" which has methods area(),volume().Define a constant PI having a value 3.142.Create a class cylinder which implements this interface (members – radius, height) Create one object and calculate the area and volume.
- d) d) Write a program to find the cube of given number using function interface.

Set B

- a) Create an abstract class "order" having members id,description.Create two subclasses "Purchase Order" and "Sales Order" having members customer name and Vendor name respectively.Define methods accept and display in all cases. Create 3 objects each of Purchase Order and Sales Order and accept and display details.
- b) Write a program to using marker interface create a class product(product_id, product_name, product_cost, product_quantity) define a default and parameterized constructor. Create objects of class product and display the contents of each object and Also display the object count.

Set C

- a) Create an interface Department containing attributes deptName and deptHead. It also has abstract methods for printing the attributes. Create a

class hostel containing hostelName, hostelLocation and numberOfRooms. The class contains method printing the attributes. Then write Student class extending the Hostel class and implementing the Department interface. This class contains attributes studentName, regNo, electiveSubject and avgMarks. Write suitable printData method for this class. Also, implement the abstract methods of the Department interface. Write a driver class to test the Student class. The program should be menu driven containing the options: i. ii. iii. Admit new student Migrate a student Display details of a student For the third option, a search is to be made on the basis of the entered registration Number

Assignment 4

Set A

a) Define a class patient (patient_name, patient_age, patient_oxy_level, patient_HRCT_report). Create an object of patient. Handle appropriate exception while patient oxygen level less than 95% and HRCT scan report greater than 10, then throw user defined Exception “Patient is Covid Positive(+) and Need to Hospitalized” otherwise display its information.

b) Write a program to read a text file “sample.txt” and display the contents of a file in reverse order and also original contents change the case (display in upper case).

c) Accept the names of two files and copy the contents of the first to the second. First file having Book name and Author name in file. Second file having the contents of First file and also add the comment ‘end of file’ at the end.

Set B

a) Write a program to read book information (bookid, bookname, bookprice, bookqty) in file “book.dat”. Write a menu driven program to perform the following operations using Random

access file: i. Search for a specific book by name. ii. Display all book and total cost

b) Define class EmailId with members ,username and password. Define default and parameterized constructors. Accept values from the command line Throw user defined exceptions – “InvalidUsernameException” or “InvalidPasswordException” if the username and password are invalid.

c) Define a class MyDate (day, month, year) with methods to accept and display a MyDate object. Accept date as dd, mm, yyyy. Throw user defined exception “InvalidDateException” if the date is invalid. Examples of invalid dates : 03 15 2019, 31 6 2000, 29 2 2021

Set C

a) Write a menu driven program to perform the following operations on a set of integers as shown in the following figure. A load operation should generate 10 random integers (2 digit) and display the number on screen. The save operation should save the number to a file “number.txt”. The short menu provides various operations and the result is displayed on the screen.

b) Write a java program to accept Employee name from the user and check whether it is valid or not. If it is not valid then throw user defined Exception “Name is Invalid” otherwise display it.(Name should contain only characters)